

MSPM0_K210 Handwritten Digit Recognition

1. K210 and MSPM0 Communication

1.1 Experimental Prerequisites

This tutorial uses the MSPM0G3519 development board. K210 needs to run the program in **K210-AI(MSPM0)** to start the experiment.

MSPM0G3519 *1

K210 vision module *1 (with SD card (with AI model) and camera)

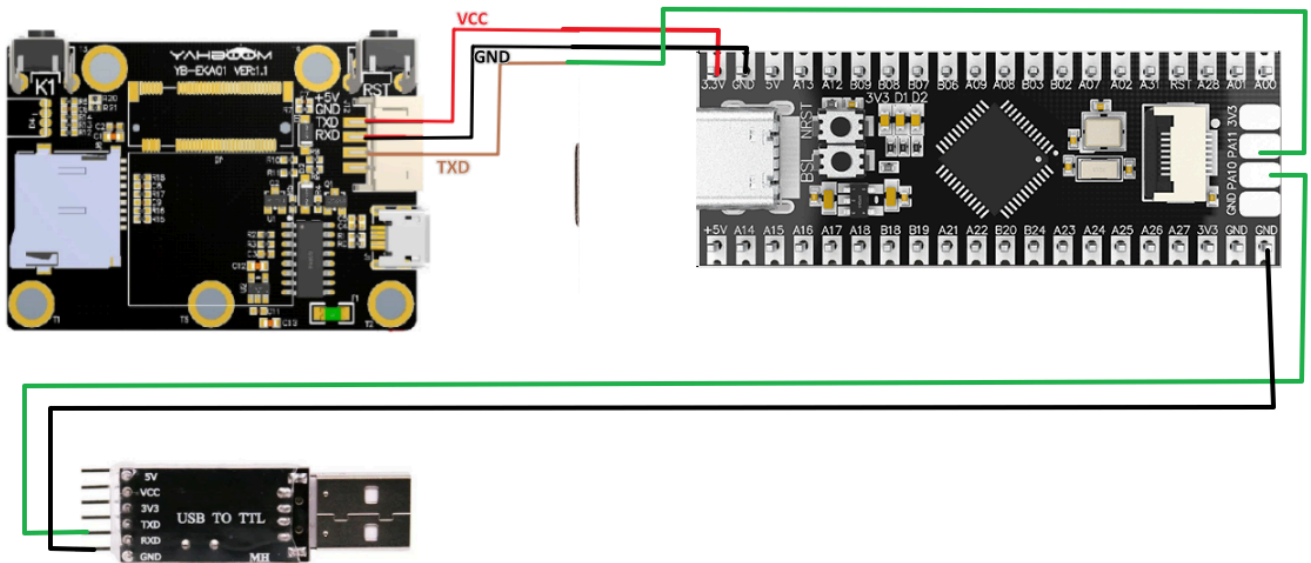
USB to TTL module *1

1.2 Experimental Wiring

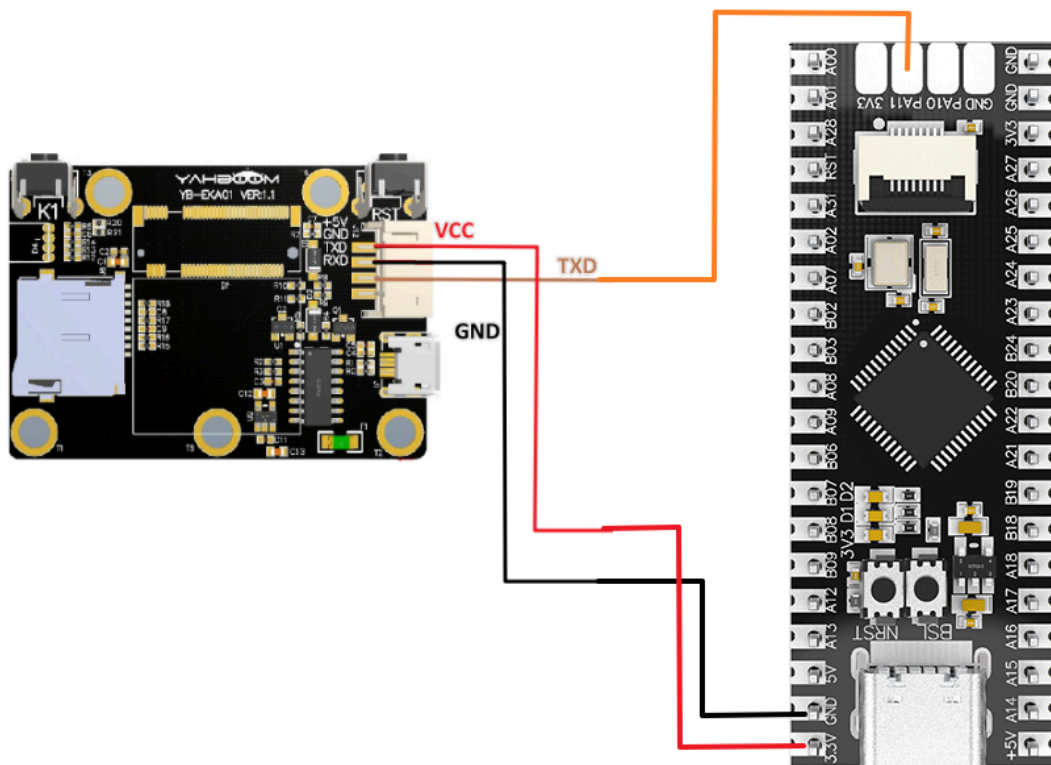
Method 1: Using USB to TTL module

MSPM0G	K210 Vision Module
RX	TXD
GND	GND
VCC	5V

MSPM0G	USB to TTL Module
TX	RXD
GND	GND



Method 2: Using onboard Type-C port



1.3 Main Code Explanation

This example sets the serial port print baud rate to 115200 bps, and the K210 module connection baud rate is also set to 115200 bps.

Type Filter Text... X <<

PROJECT CONFIGURATIO...
Project Config... 1/1 ✓ +

MSPM0 DRIVER LIBRARY ...

SYSTEM (9)
Board 1/1 ✓ +
DMA +
GPIO +
MATHACL +
Configuration NVM +
RTC +
SYSCTL 1/1 ✓ +
SYSTICK 1/1 ✓ +
WWDT +

ANALOG (6)
ADC12 +
COMP +
DAC12 +
GPAMP +
OPA +
VREF +

COMMUNICATIONS (6)
I2C +
I2C - SMBUS +
MCAN +
SPI +
UART 2/4 ✓ +
UART - LIN +

TIMERS (6)
TIMER - CAPTURE +

Software > UART

UART (2 of 4 Added) ⓘ
+ ADD REMOVE ALL

✓ MYUART
✓ K210_Uart

Name MYUART

Selected Peripheral UART0

Quick Profiles

UART Profiles Custom

Basic Configuration

UART Initialization Configuration

Clock Source BUSCLK

Clock Divider Divide by 1

Calculated Clock Source 32.00 MHz

Target Baud Rate 115200

Calculated Baud Rate 115211.52

Calculated Error (%) 0.01

Word Length 8 bits

Parity None

Stop Bits One

HW Flow Control Disable HW flow control

Type Filter Text...
X
<<
<
>
Software > UART

PROJECT CONFIGURATION...

Project Config... 1/1 ✓ (+)

MSPM0 DRIVER LIBRARY ...

SYSTEM (9)

Board 1/1 ✓ (+)

DMA (+)

GPIO (+)

MATHACL (+)

Configuration NVM (+)

RTC (+)

SYSCTL 1/1 ✓ (+)

SYSTICK 1/1 ✓ (+)

WWDT (+)

ANALOG (6)

ADC12 (+)

COMP (+)

DAC12 (+)

GPAMP (+)

OPA (+)

VREF (+)

COMMUNICATIONS (6)

I2C (+)

I2C - SMBUS (+)

MCAN (+)

SPI (+)

UART 2/4 ✓ (+)

UART - LIN (+)

TIMERS (6)

TIMER - CAPTURE (+)

TIMER - COMPARE (+)

UART (2 of 4 Added) ②

✓ MYUART

✓ K210_Uart

Name

K210_Uart

Selected Peripheral

UART1

Quick Profiles

UART Profiles

Custom

Basic Configuration

UART Initialization Configuration

Clock Source

BUSCLK

Clock Divider

Divide by 1

Calculated Clock Source

32.00 MHz

Target Baud Rate

115200

Calculated Baud Rate

115211.52

Calculated Error (%)

0.01

Word Length

8 bits

Parity

None

Stop Bits

One

HW Flow Control

Disable HW flow control

```

int main(void)
{
    SYSCFG_DL_init();

    NVIC_ClearPendingIRQ(MYUART_INST_INT_IRQN); //Clear serial port interrupt flag
    NVIC_EnableIRQ(MYUART_INST_INT_IRQN); //Enable serial port interrupt

    while (1)
    {
        if (k210_msg.class_n != 0) //Example number is not empty
        {
            if(k210_msg.class_n == 11) //Handwritten digit recognition
            {
                sprintf(buff_com, "class =
%d,x=%d,y=%d,w=%d,h=%d\r\n", k210_msg.class_n, k210_msg.x, k210_msg.y, k210_msg.w, k210_msg.h);
                uart0_send_string(buff_com);
                k210_msg.class_n = 0;
            }
        }
    }
}

```

```

    }

    }
    delay_ms(500);

}
}

```

After the above program runs this example, the k210_msg structure members will have corresponding values and will be printed via serial port.

k210_msg: is the structure for receiving information. Its main members are:

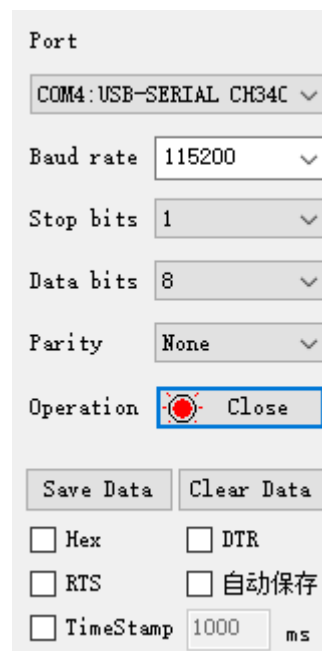
- x: is the x-coordinate of the upper-left corner of the recognized box (range: 0-240)
- y: is the y-coordinate of the upper-left corner of the recognized box (range: 0-320)
- w: is the width of the recognized box (range: 0-240)
- h: is the height of the recognized box (range: 0-320)
- id: is the recognized label
- class_n: example number
- msg_msg[20]: valid data

After data reception and processing, each member of k210_msg will store valid information. If you want to do secondary development, you can directly call the members of k210_msg.

Note: The Keil project source code must be placed under the SDK path for compilation.

1.4 Experimental Phenomenon

1. After connecting the cables, the K210 vision module runs offline [K210 offline running method](#)
2. Set the serial assistant interface as shown in the figure



3. Then run the handwritten digit recognition example. The serial assistant will print the important information transmitted by K210 to MSPM0G, as shown in the figure below.

```
id = 0
id = 8
id = 8
id = 8
id = 8
id = 8
id = 8
id = 8
id = 8
id = 8
id = 8
id = 8
id = 8
id = 8
id = 8
id = 0
id = 0
id = 6
id = 6
id = 5
id = 1
id = 9
id = 9
id = 1
id = 5
id = 6
id = 0
id = 6
id = 5
```

Handwritten digit recognition only transmits the 1 member variable of k210_msg: id.

id: is the recognized digit ID (0-9)